RewardDS: Privacy-Preserving Fine-Tuning for Large Language Models via Reward Driven Data Synthesis

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Abstract

The success of large language models (LLMs) has attracted many individuals to fine-tune them for domain-specific tasks by uploading their data. However, in sensitive areas like healthcare and finance, privacy concerns often arise. One promising solution is to generate synthetic data with Differential Privacy (DP) guarantees to replace private data. However, these synthetic data contain significant flawed data, which are considered as noise. Existing solutions typically rely on naive filtering by comparing ROUGE-L scores or embedding similarities, which are ineffective in addressing the noise. To address this issue, we propose *RewardDS*, a novel privacy-preserving framework that fine-tunes a reward proxy model and uses reward signals to guide the synthetic data generation. Our *RewardDS* introduces two key modules, Reward Guided Filtering and Self-Optimizing Refinement, to both filter and refine the synthetic data, effectively mitigating the noise. Extensive experiments across medical, financial, and code generation domains demonstrate the effectiveness of our method. Our code and data will be available at https://github.com/wjw136/RewardDS.

1 Introduction

The remarkable capabilities of Large Language Models (LLMs) in general tasks have motivated many individuals and organizations to customize their own LLMs for domain-specific applications, such as medical diagnosis, financial analysis, etc. (Wu et al., 2023; Chen et al., 2023). While domain adaptation through fine-tuning is attractive, high computational costs make local fine-tuning impractical for most users. Currently, most LLM service providers (Achiam et al., 2023; Yang et al., 2024a; Doubao, 2024) offer fine-tuning services, allowing users to customize LLMs for their needs by preparing and uploading their domain-specific data. However, these data may contain sensitive information, and directly transferring it to the LLM service provider can lead to significant privacy concerns (Zeng et al., 2024; Abdelnabi et al., 2023). We denote the individuals or organizations which aim to customize LLMs as **client**, the LLM service providers as **server**, and the model being fine-tuned as the **target LLM**. It remains a critical challenge to develop privacy-preserving fine-tuning methods in such a client-server scenario.

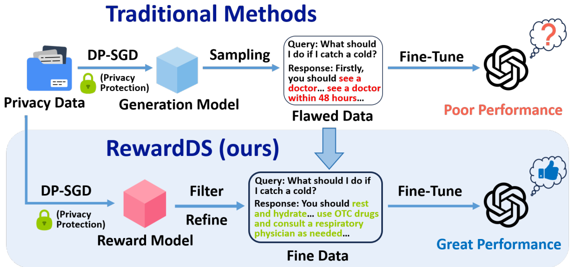

Figure 1: Illustration of how *RewardDS* overcomes the dilemma of traditional synthetic data methods. The synthetic data directly sampled from the generation proxy model contain significant flaws, such as incoherent text or incomplete storylines, which are considered noise.

Prior works proposed data synthesis as a promising solution (Kurakin et al., 2023; Yue et al., 2023; Yu et al., 2023; Mattern et al., 2022; Flemings and Annavaram, 2024). These approaches generate synthetic data to replace the private data and can be used for fine-tuning, thus ensuring privacy protection. Specifically, a generation proxy model is first trained on the private data, optimized by DP-SGD (Abadi et al., 2016) to safeguard privacy. The generation proxy model then generates synthetic data for subsequent LLM training. However, due to the inherent randomness of the generation process, the synthetic data inevitably contain significant flawed one, including text incoherence or storyline incompleteness, which is considered as noise and leads to less effective LLM fine-tuning, as illustrated in Figure 1.

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To mitigate the noise, existing methods (Yu et al., 2024; Wang et al., 2022; Xie et al., 2024) proposed to filter out flawed data by measuring its similarity to private data. Wang et al. (2022) use ROUGE-L similarity, while Yu et al. (2024); Xie et al. (2024) compute embedding similarity. However, these metrics fail to evaluate the synthetic data's effectiveness for domain-specific tasks. Alternative methods (Wang et al., 2024; Li et al., 2024b) attempt to distill the capabilities from the LLM on the server side into the generation proxy model to support domain-specific tasks. However, since the LLM is not fine-tuned for these specific domains, the distillation provides limited benefit and does not effectively improve task performance. We also present an illustrative example in Figure 7.

To more effectively mitigate noise in synthetic data, we propose *RewardDS* (**Reward**-driven **D**ata **S**ynthesis), a novel privacy-preserving framework that improves synthetic data quality for the target LLM's privacy-preserving fine-tuning. RewardDS implements a two-stage quality control process, i.e., filtering and refinement, as illustrated in Figure 1. Specifically, we first train a reward proxy model on private data to assess data quality for domain-specific tasks, using DP-SGD to safeguard privacy. Through **Reward Guided Filtering**, we apply the reward proxy model to assess synthetic data generated by the generation proxy model and remove samples with low reward scores. Filtering alone may remove a large amount of data, leaving only a small fraction. Therefore, we aim to further refine the synthetic data to obtain more high-quality data. Our **Self-Optimizing Refinement** module generates multiple candidate responses for each synthetic query and computes their rewards. The generation proxy model analyzes the highest and lowest scoring responses and then generates improvement feedback. Based on this feedback, the target LLM refines the synthetic data following a refinement instruction. The resulting high-quality, filtered, and refined synthetic data are then used to fine-tune the target LLM for domain-specific tasks.

We conduct extensive experiments across various domain-specific generation tasks, including Medical Question Answering (QA), Legal QA, and Code Generation tasks. The results consistently demonstrate the effectiveness of our method in improving the quality of the synthetic data, achieving better performance while preserving privacy. Our main contributions are summarized as follows:

- We propose *RewardDS*, a novel privacy-preserving fine-tuning framework that improves the quality of synthetic data by training a Reward Proxy Model on the client side to guide synthetic data generation.
- We introduce the Reward Guided Filtering and Self-Optimizing Refinement modules to filter and refine the synthetic data, thereby enhancing its quality.
- We conducted extensive experiments across Medical QA, Legal QA, and Code Generation tasks to validate the effectiveness of our proposed framework.

2 Related Work

In this section, we will introduce the related work on LLM privacy-preserving fine-tuning methods, which are currently divided into three categories: Anonymity-based methods, Encryption-based methods and Synthesis-based methods.

Anonymity-based methods. Techniques like k-anonymity and adversarial anonymization can identify and anonymize private data. But they will significantly harm data quality for domain-specific LLM fine-tuning on the server side. (Staab et al., 2024; Sweeney, 1997; Romanov et al., 2019)

Encryption-based methods. Some approaches employ encryption techniques, such as Homomorphic Encryption (HE) or Secure Multi-Party Computation (SMPC), to protect private data. However, encrypting data and maintaining secure communication between server and client incur significant computational and time overhead, making these methods impractical in real-world scenarios. (Frery et al., 2025; Lou et al., 2020; You et al., 2025)

Synthesis-based methods. Recent studies have explored using synthetic data with differential privacy (DP) guarantees as a substitute for private data in LLM fine-tuning. While this offers a practical and efficient solution, the synthetic data often contain noisy or flawed samples that significantly hinder LLM fine-tuning. Simple filtering based on text similarity is insufficient to effectively eliminate such noise. (Kurakin et al., 2023; Yue et al., 2023; Yu et al., 2024; Hou et al., 2024; Wang et al., 2024, 2022)

Due to the limited space, a detailed introduction of the above works can be found in Appendix A.

3 Problem Statement

We consider a scenario where the client holds domain-specific data, such as patient's medical records, which contain sensitive information. Hence, directly transmitting those data to servers for LLM fine-tuning is not allowed. This private data typically is structured as *Query-Response* pairs, with both query and response containing confidential private information (Wang et al., 2024). The server, which hosts the target LLM, offers only API access while keeping model weights confidential, preventing clients from accessing or locally fine-tuning the model. While clients can fine-tune some lightweight LLMs within their computational constraints, these models have inherently weaker capabilities than the target LLM. This creates a critical challenge: how to leverage a client's private data to improve the server-hosted LLM's performance on domain-specific tasks while preserving privacy.

Existing synthesis-based methods utilize a lightweight **Generation Proxy Model** to generate safe synthetic data for fine-tuning the target LLM on the server (Yue et al., 2023; Yu et al., 2024). However, the randomness of the generation process introduces significant noise in the synthetic data, potentially causing performance degradation. Therefore, our main goal is to ***explore a more effective method for mitigating the noise in synthetic data, enabling better fine-tuning performance while maintaining user privacy***.

4 Method

To address the performance degradation caused by noise in synthetic data, we propose a novel framework, *RewardDS* (**Reward**-driven **D**ata **S**ynthesis), as shown in Figure 2. Our approach additionally trains a **Reward Proxy Model** on the client side. Then the reward proxy model filters and refines the synthetic data sampled from the generation proxy model through **Reward Guided Filtering** and **Self-Optimizing Refinement** modules on the server side. Both modules collaborate to enhance the quality of the synthetic data, driven by the reward signal from the reward model. We will introduce the training process of the generation proxy model and reward proxy model in § 4.1. The details of reward guided filtering and self-optimizing refinement module are provided in § 4.2.

4.1 Client Side

Generation Proxy Model Training. The generation proxy model is responsible for generating safe synthetic data as a substitute for private data. Following (Yue et al., 2023; Yu et al., 2024, 2022; Kurakin et al., 2023), we fine-tune a generation proxy model on the client's private data using the DP-SGD algorithm (Abadi et al., 2016). The backbone of generation proxy model should be lightweight due to limited computational resources on the client side, e.g., Qwen2.5-0.5B-Instruct (Yang et al., 2024b). The DP-SGD algorithm protects the privacy of the training data by injecting noise into the gradients during model training. This noise ensures that the inclusion or exclusion of any individual training sample has minimal impact on the fine-tuned model, thereby providing privacy protection.

Reward Proxy Model Training. The reward model is responsible for evaluating the quality of the synthetic data. It should provide higher rewards for high-quality data while lower rewards for poor-quality data. Following standard reward model training practices (Liu et al., 2024), we train the reward proxy model using paired comparison data. Let W_0 denote the initial backbone model, W_{gen} the fine-tuned generation proxy model, and W_{rwd} the fine-tuned reward proxy model. For each query Q from the private dataset with its gold response A_{gold}, we generate two responses: A_0 from W_0 and A_{gen} from W_{gen}. We then create preference pairs by selecting either A_{gen} or A_{gold} as the chosen response A_c, with A_0 serving as the rejected response A_r. The reward proxy model maintains a lightweight architecture for client-side deployment and is fine-tuned using differential privacy (DP-SGD) to prevent privacy leakage.

Following Rafailov et al. (2023), we define the training loss as:

$$\mathcal{L} = -\log \sigma \left(f_{\text{rwd}}(Q, A_c) - f_{\text{rwd}}(Q, A_r) \right), \quad (1)$$

where $f_{\text{rwd}(\cdot)}$ represents the reward predicted by W_{rwd}. This training loss encourages the reward model to assign higher scores to responses from the generation proxy model and gold responses compared to those from the initial backbone model.

After training, both generation proxy model and reward proxy model are sent to the server.